Owen Pearson

California Polytechnic State University San Luis Obispo

4th Year B.S. Electrical Engineering

Graduating Spring 2027



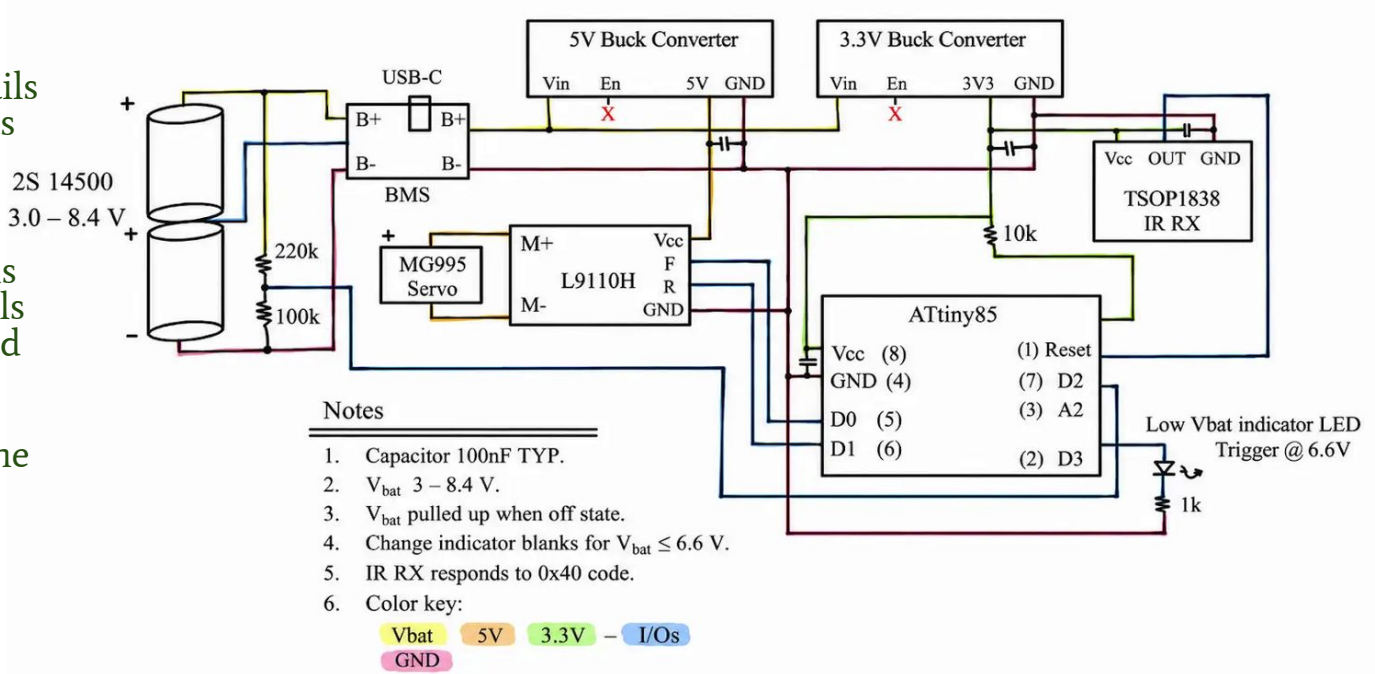
Personal: (In Development) Remote Operated Rolling Shade

There is a large glass door in my bedroom that the sun shines through each morning. To sleep in or rest during the day, I typically hang a towel over the door. This remote operated rolling shade is battery powered and can be controlled from my bedside. An embedded ATtiny85 is used to process remote signals, operate the H-bridge to control servo rotation, and determine the Li-ion battery SOC using a voltage divider and ADC input.

Buck converters are used to establish steady 3.3 and 5 volt rails for the ATtiny85 and continuous servo which rotates the shade, respectively.

A 2S BMS with a USB-C input is included with the two 14500 cells which can easily be disconnected for recharging.

The ADC is sample each time the motor stops spinning. The associated LED will flash if the battery needs to be charged (measurement <= 6.6V).



Personal: Auto-Filling Filtered Water Storage

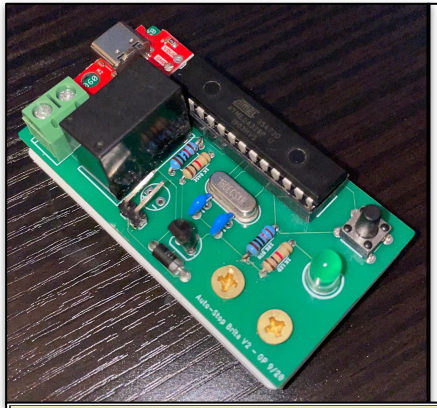
Living off-campus, filtered water either needs to be bought in single use plastic bottles, or stored in 5 gallon jugs which need to be refilled at local grocery stores. Brita water filtration systems are very popular among college students, especially those living in dorms. This project eliminates the time and hassle associated with filling a Brita water tank each time it runs low.

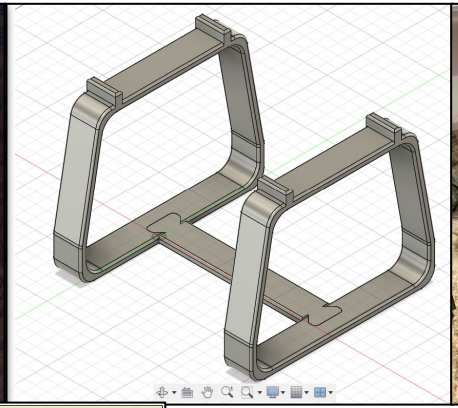
By utilizing water lines connected to a sink faucet or fridge, water is routed to the top of the tank.

Using a push button to trigger a relay-operated solenoid valve, water begins to flow until a floating reed switch within the tank closes.

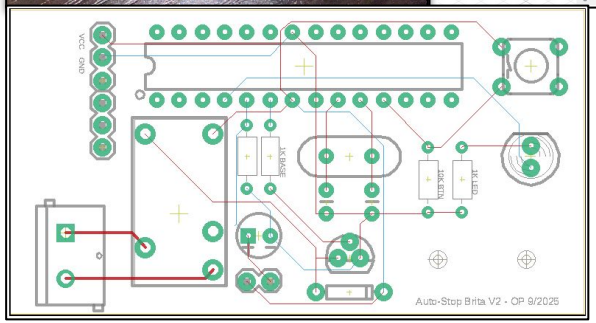
An embedded ATmega328p chip is programmed to operate the relay coil and LED indicator light.

3D printed legs elevate the tank from the countertop so that cups and bottles can be filled easily.









Above: Complete PCB, and final product.

Left: Eagle PCB layout and traces.

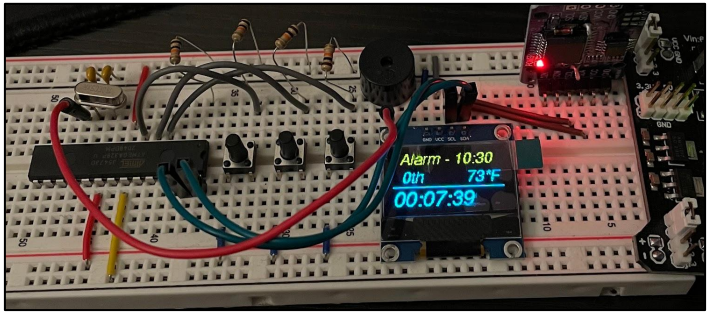
Personal: OLED Display Alarm Clock

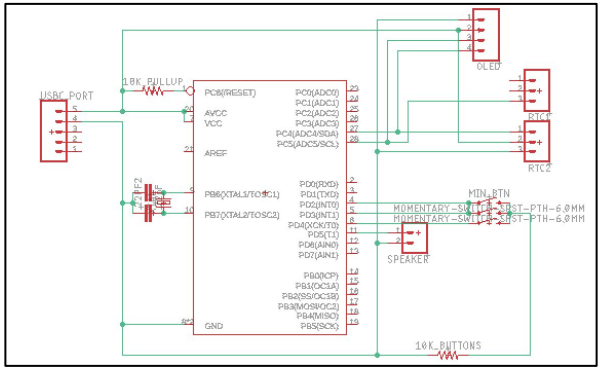
Keeping alarms on my phone next to my bed inevitably leads to distractions, as it's the first thing I interact with after a night's sleep. In an effort to minimize morning screen-time, I've began to design an alarm clock using an ATmega328p and OLED display. Time is kept by a real-time-clock module (RTC) - which has a backup battery to keep time even if power is lost. The three buttons are used to toggle the alarm on/off, hours, and minutes.

The next big step of this project is setting up the 328p chip to be programmed while isolated from its companion Arduino Uno R3.

After this is complete, the month, day, hours, minutes, and seconds of the RTC can be programmed accurately.

"st," "nd," "rd," and "th" are added to date values, and leap years are accounted for by the RTC. The RTC also gives a temperature readout which is included on the OLED.





CURRENT_OLED_RTC.ino

```
}

void printDate() {
  char *months[] = {" ", "Jan", "Feb", "Mar", "Apr", "May", "Jun",
  "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
  display.setCursor(0, 32);
  display.setTextSize(1);
  display.setFont(&FreeSansOblique9pt7b);
  display.print(months[Mon]);
  display.print(" ");
  display.print(Day);

  if (Day == 11 | Day == 12 | Day == 13) {
    display.print("th");
  }
  else {
    if(Day % 10 == 1) {
      display.print("st");
    }
    if (Day % 10 == 2) {
      display.print("nd");
    }
    if (Day % 10 == 3) {
      display.print("rd");
    }
    else {
      display.print("th");
    }
  }
```

Above: Code segment from Arduino IDE

Left: Eagle schematic for developing PCB design.

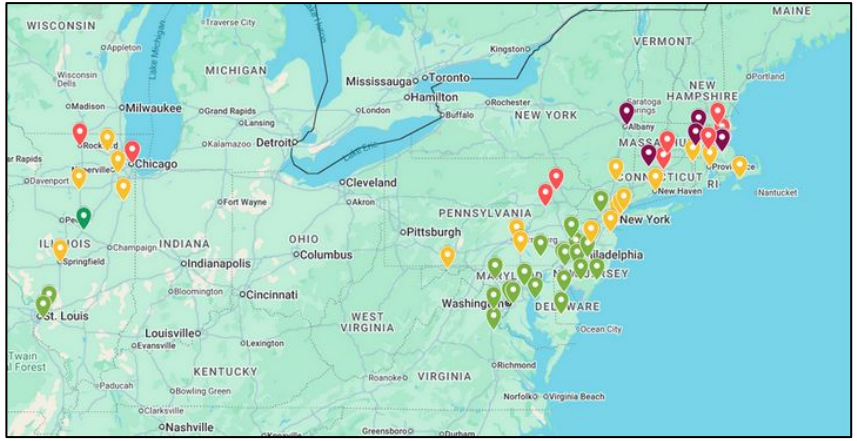
Professional: Solar Production Loss Calculator

Mar 2025 - June 2025

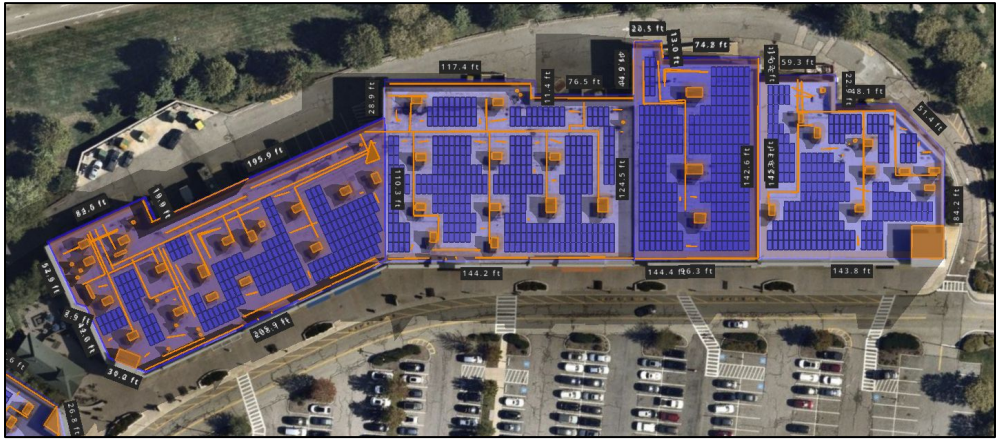
KING ENERGY

For the summer of 2025, I worked full-time as an Energy System Design Specialist at King Energy. King Energy specializes in multi-tenant commercial solar, in which the energy from a solar program must be shared across many individual businesses. The bulk of the companies' project scope is in New England, where winter storms accumulate snow on panels and limit production. Modeling these losses accurately is necessary when seeking funding from finance partners. However, existing weather modeling tools in the solar industry, such as Solcast and Solar Resource Compass, only provide data for the weather station nearest to a given address.

By collecting weather data for all prospective project locations and additional cities throughout the East Coast, zonal averages of annual snowfall could be created. From there, a Google Apps Script could be written to provide the nearest weather data and zonal averages given an input address.



Zonal snow loss categories, colored by severity.



Sample solar layout in Helioscope CAD software.

Professional: Solar Production Loss Calculator (continued)

Mar 2025 - June 2025

Below is the final interface for the snow-loss tool. Given the address of a prospective project location:

- The nearest four locations with weather data are returned, along with their distance and cardinal direction.
- The zonal average corresponding to the input address is highlighted.
- Production losses for a 1-megawatt project geographically centered on the East Coast are provided.
  - Losses are calculated via the nearest weather station or zonal average.
  - Losses may be displayed as a percentage or kWh quantity.

Address		RUN	< Click	Existing SRC Data Map	1MW Parsippany, NJ Helio
10600 Belmont Ave, Franklin Park, IL 60131				Map	Helio

SRC				% Loss due to Snow												% Loss due to Dust											
Nearest SRC Report	Zone	Distance (mi)	Direction	Monthly Avg	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Monthly Avg	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Chicago, IL (city center)	4	13.6	East	6.5	24.0	20.0	12.0	4.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	15.0	0.7	1.0	1.0	1.0	1.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0	0.0
Next 4 Nearest SRC Reports																													
2100 Internationale Pky Woodridge, IL 60517	3	18.5	Southwest	5.7	21.0	18.0	10.0	3.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	14.0	0.7	1.0	1.0	1.0	1.0	0.0	0.0	1.0	0.0	1.0	1.0	1.0	0.0
3031 Broadsmore Dr Algonquin, IL 60102	3	26.8	Northwest	5.6	20.0	16.0	8.0	3.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	17.0	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0	1.0	1.0	1.0	1.0
Kankakee, IL (city center)	3	56.6	South	4.2	19.0	10.0	6.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	12.0	0.6	0.0	1.0	1.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0	1.0	1.0
Rockford, IL (city center)	4	65.6	West	6.4	23.0	17.0	10.0	3.0	0.0	0.0	0.0	0.0	0.0	0.0	5.0	19.0	0.7	1.0	1.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0
Next 4 Nearest SRC Reports **Average**																													
				5.5	20.8	15.3	8.5	2.8	0.0	0.0	0.0	0.0	0.0	0.0	2.8	15.5	0.7	0.8	1.0	1.0	0.8	0.5	0.3	0.5	0.3	1.0	1.0	1.0	0.5
Zone **Average**																													
	1			0.8	3.5	1.5	1.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.0	0.8	1.0	1.0	0.5	1.0	0.5	0.5	0.5	0.5	1.0	1.0	1.0	1.0
	2			2.8	10.7	11.1	5.4	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	5.4	0.8	1.0	1.0	0.9	0.9	0.7	0.8	0.4	0.5	1.0	1.0	1.0	1.0
	3			4.7	17.3	15.2	9.1	2.7	0.0	0.0	0.0	0.0	0.0	0.0	1.2	11.0	0.7	0.8	0.9	0.7	0.6	0.7	0.4	0.6	0.4	0.7	0.8	0.7	0.6
	4			7.0	24.0	20.6	13.5	5.0	0.5	0.0	0.0	0.0	0.0	0.4	3.3	16.3	0.6	0.8	0.9	0.5	0.6	0.5	0.3	0.6	0.4	0.7	0.4	0.6	0.3
	5			8.8	29.2	24.7	16.7	7.0	1.0	0.0	0.0	0.0	0.0	0.5	4.8	22.3	0.8	0.8	0.8	0.7	0.7	0.8	0.8	1.0	1.0	1.0	0.8	0.8	0.7

Click > **TRIGGER** (kWh vs. % Losses)

Nearest SRC Report	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
Production (% / amount)	54,818	72,947	114,075	133,950	152,822	156,050	161,414	147,078	120,555	84,643	58,823	47,373	Snow Production Losses	4.3%	55663.2 kWh
w/ Snow Losses	41,662	58,357	100,386	128,592	152,822	156,050	161,414	147,078	120,555	84,643	57,059	40,267	Dust Production Losses	0.6%	7526.7 kWh
w/ Dust Losses	41,245	57,774	99,382	127,306	152,822	156,050	159,800	147,078	119,350	83,796	56,488	40,267	Total Production Losses	4.8%	63189.8 kWh

Zone Average	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
Production (% / amount)	54,818	72,947	114,075	133,950	152,822	156,050	161,414	147,078	120,555	84,643	58,823	47,373	Snow Production Losses	4.7%	61046.4 kWh
w/ Snow Losses	41,662	57,920	98,675	127,252	152,058	156,050	161,414	147,078	120,555	84,304	56,882	39,651	Dust Production Losses	0.5%	6538 kWh
w/ Dust Losses	41,328	57,398	98,181	126,489	151,298	155,582	160,446	146,489	119,711	83,967	56,541	39,532	Total Production Losses	5.2%	67584.5 kWh

Map Data

INPUT

** Enter the full project address in cell B3.

** Press the "Run" button in cell D3.

OUTPUT

** Monthly snow & dust soiling losses for the nearest SRC project.

** Individual montly snow & dust soiling losses for the next four nearest SRC projects.

** An AVERAGE of monthly snow & dust soiling losses across the next four nearest SRC projects to the address of the project.

** Monthly average snow & dust losses for the "Zone" the project address falls within. (Highlighted Green)

** Monthly production as a percent / amount of total output for a 1MW East Coast Project.

** Production loss projections for snow & dust for both the nearest SRC project and the Zone average of the project (with toggle for amount /

Lab Design Project: Sun Tracking “Solar Panel”

In order to maximize the amount of solar energy captured in a day, solar panels are often rotated along one (or two) axis to follow the sun. In this design, by comparing the brightness on one side of the panel to another, its tilt can be adjusted until the two brightnesses reach an equilibrium.

- Two LDR’s on opposing sides of the panel provide varying voltage levels for comparison.
- If compared values differ by more than 10%, the servo motor is operated to adjust tilt angle.
- Tilt angle is mapped to an approximate time of day.
- Before “sunrise” and after “sunset,” the LCD display indicates whether the homes battery is charging or discharging. During nighttime hours, lighting within the home is automatically switched on.
- A hall-effect sensor is used as a dimmer-switch to vary the current applied to the LED lighting. Brightness is controlled by the distance of a magnet from the sensor.

Left: Close-up image of hall-effect dimmer switch.

Right: Final Product.

[Demo Link]

